EDF optimality: proof (1/3)

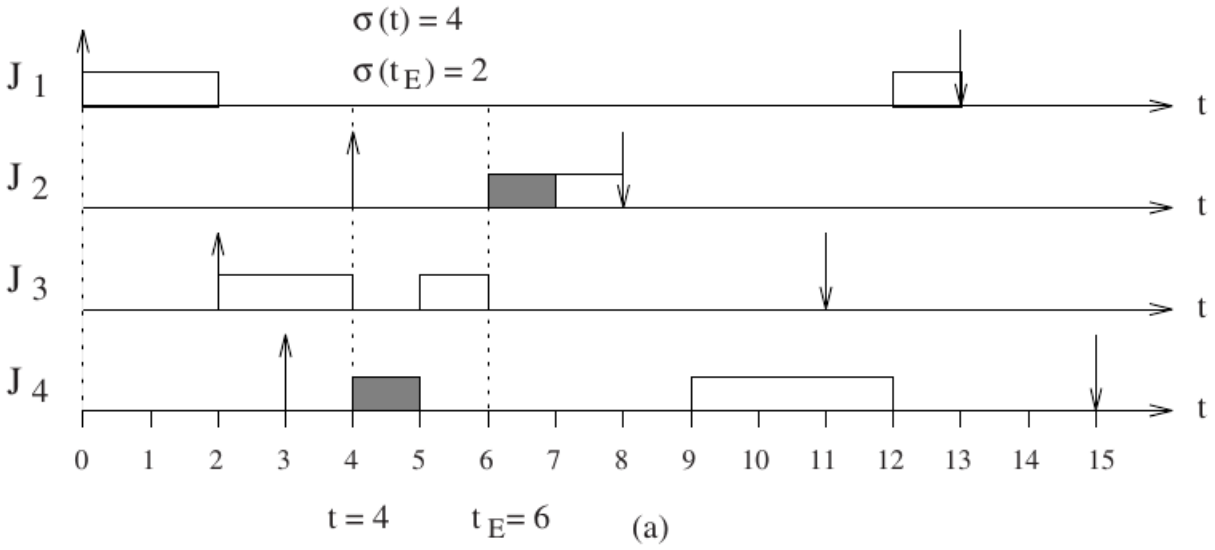
- A feasible schedule σ for task set Γ is divided into time slices of one time unit (note that the time slice duration could be arbitrary small):
  - $\sigma(t)$:= task executing during the time slice $[t, t+1)$
  - $E(t)$:= index of the task with minimum absolute deadline at t
  - t_E := time $\geq t$ at which $\tau_{E(t)}$ is executed first
  - $d_{\max} := \max_{i \in \{1,\ldots,n\}} \{d_i\}$ (maximum absolute deadline)
- Schedule σ is transformed into an EDF schedule σ_{EDF}:

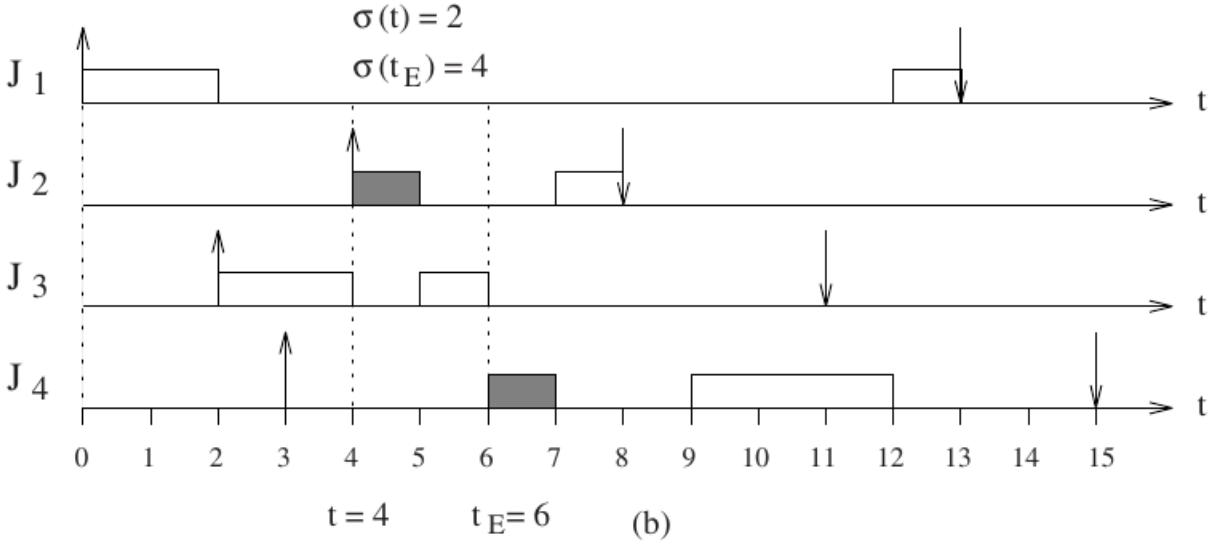
input : A feasible schedule σ for task set Γ
output : An EDF schedule σ_{EDF} for task set Γ

foreach $t \in \{0, 1, \ldots, d_{\max} - 1\}$ **do**
  if $\sigma(t) \neq \sigma_{\text{EDF}}(t)$ **then**
    $\sigma(t_E) = \sigma(t)$ // $[t_E, t_E + 1)$ allocated to the task executed during $[t, t+1)$
    $\sigma(t) = E(t)$ // $[t, t+1)$ allocated to the task with min absolute deadline at t
  end
end
return TRUE

EDF optimality: proof (2/3)

ESEMPIO DI SCAMBIO





Images from "Hard real-time computing systems" by Prof. G. Buttazzo

EDF optimality: proof (3/3)

- A transposition preserves schedulability, i.e., σ_{EDF} is feasible for Γ
  - If a time slice of task τ_i is anticipated $\Rightarrow$ The feasibility of τ_i is preserved
  - If a time slice of task τ_i is postponed at t_E $\Rightarrow$

    $\Rightarrow t_E + 1 \leq d_E$ with $d_E :=$ earliest absolute deadline at t, since σ is feasible $\Rightarrow$

    $\Rightarrow t_E + 1 \leq d_E \leq d_i$ $\forall$ task τ_i by definition of d_E $\Rightarrow$

    $\Rightarrow$ the time slice postponed at t_E is schedulable

OVVERO:

PER HP LO SCHEDULE È FASIBILE E NON EDF. FACENDO SI DIVENTA EDF ED È ANCORA FEASIBLE PERCHÉ: IL JOB CHE VIENE ANTICIPATO NON AVRÀ PROBLEMI; QUELLO CHE VIENE POSTICIPATO NON HA PROBLEMI PERCHÉ QUELLO CHE È STATO ANTICIPATO NON AVEVA PROBLEMI E AVEVA LA DEADLINE PRIMA DI QUELLO CHE È STATO POSTICIPATO.